YEAR 9 MATHS · CHAPTER 2 · MIND MAP

# Expressions and linear equations

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## 1 Algebraic expressions

A pronumeral stands for a number. $3x$ means $3 \times x$.

"5 more than $x$": $x + 5$  "twice $y$, minus 4": $2y - 4$

If $a = 3,\ b = -2$: $2a - b^2 = 6 - 4 = 2$

**Watch out:** substitute negatives in brackets: $(-2)^2 = 4$

## 2 Simplifying expressions

Only like terms (same letters, same powers) can be added or subtracted.

$4x + 3y - x + 2y = 3x + 5y$

$3a \times 4ab = 12a^2b$  $12xy \div 4x = 3y$

**Watch out:** $x + x = 2x$ but $x \times x = x^2$

## 3 Expanding expressions

Multiply every term inside the brackets by the term outside.

$3(x + 4) = 3x + 12$  $x(x + 7) = x^2 + 7x$

$2(x + 3) - (x - 1) = 2x + 6 - x + 1 = x + 7$

**Watch out:** a minus sign outside changes every sign inside.

## 4 Solving equations: one side

Undo the operations in reverse order. Do the same to both sides.

$3x + 5 = 20 \Rightarrow 3x = 15 \Rightarrow x = 5$

$\frac{x}{4} - 2 = 3 \Rightarrow \frac{x}{4} = 5 \Rightarrow x = 20$

Check by substituting: $3(5) + 5 = 20$ ✓

## 5 Brackets and pronumerals on both sides

Expand brackets first, then collect the $x$ terms on one side.

$2(x - 3) = 10 \Rightarrow 2x - 6 = 10 \Rightarrow x = 8$

$5x + 2 = 2x + 14 \Rightarrow 3x = 12 \Rightarrow x = 4$

$4(x + 1) = 2(x + 7) \Rightarrow 4x + 4 = 2x + 14 \Rightarrow x = 5$

**Watch out:** subtract the smaller x term, so the x stays positive.

## 6 Solving word problems

Let $x$ be the unknown, write an equation, solve it, then answer in words.

"Triple a number, add 7, get 31": $3x + 7 = 31$, so the number is 8

Rectangle 3 cm longer than it is wide, perimeter 26 cm: $2(w + w + 3) = 26 \Rightarrow w = 5$, so 5 cm by 8 cm

## 7 Solving linear inequalities

Solve like an equation, but flip the sign when you multiply or divide by a negative.

$2x + 3 < 11 \Rightarrow 2x < 8 \Rightarrow x < 4$

$4 - 2x > 10 \Rightarrow -2x > 6 \Rightarrow x < -3$

On a number line: $<$ or $>$ is an open circle, $\leq$ or $\geq$ is a closed dot.

## 8 Using formulas

Substitute the values you know, then solve for the one that's left.

$v = u + at$ with $u = 5,\ a = 2,\ t = 3$: $v = 11$

$A = \frac{1}{2}bh$ with $A = 24,\ b = 8$: $24 = 4h$, so $h = 6$

Rearrange for $t$: $t = \dfrac{v - u}{a}$

## 9 Simultaneous equations: substitution

Replace one pronumeral with what it equals from the other equation.

$y = 2x$ and $x + y = 12$

$x + 2x = 12 \Rightarrow x = 4$, then $y = 8$

Check in both equations: 8 = 2(4) ✓ and 4 + 8 = 12 ✓

## 10 Simultaneous equations: elimination

Add or subtract the equations to remove one pronumeral.

$x + y = 10,\ x - y = 4$: add, $2x = 14$, so $x = 7,\ y = 3$

$2x + 3y = 13,\ 2x + y = 7$: subtract, $2y = 6$, so $y = 3,\ x = 2$

## 11 Applications of simultaneous equations

Two unknowns need two equations. Use a letter for each unknown.

Adult tickets $15, child tickets $8. 7 tickets cost $77.

$a + c = 7$ and $15a + 8c = 77$

Substitute $a = 7 - c$: $105 - 7c = 77$, so $c = 4$ and $a = 3$: 3 adults and 4 children